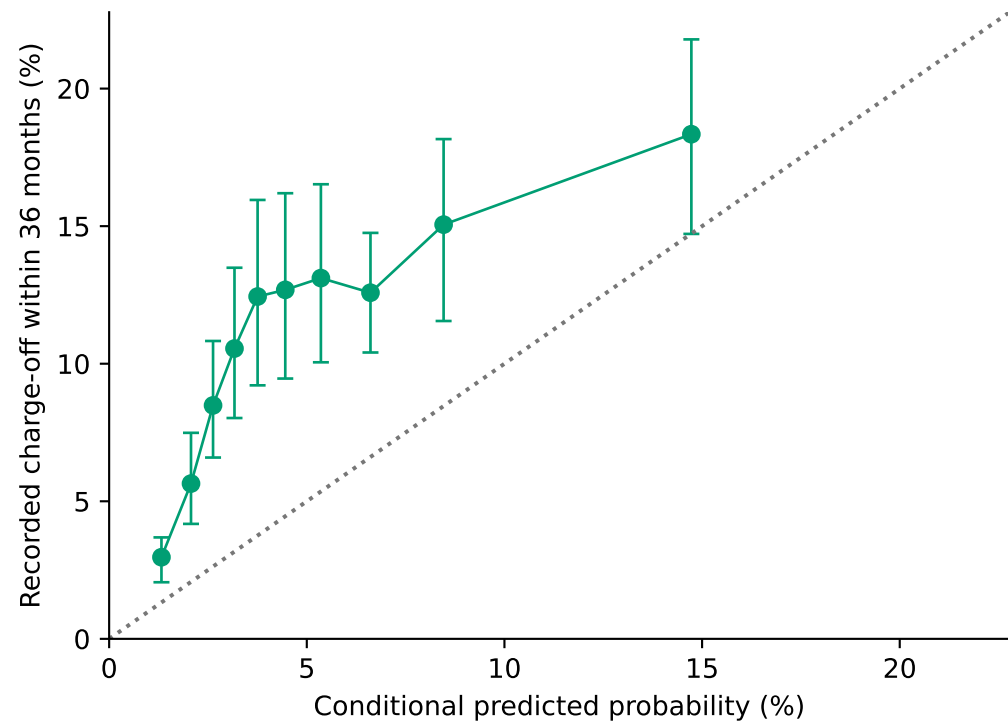
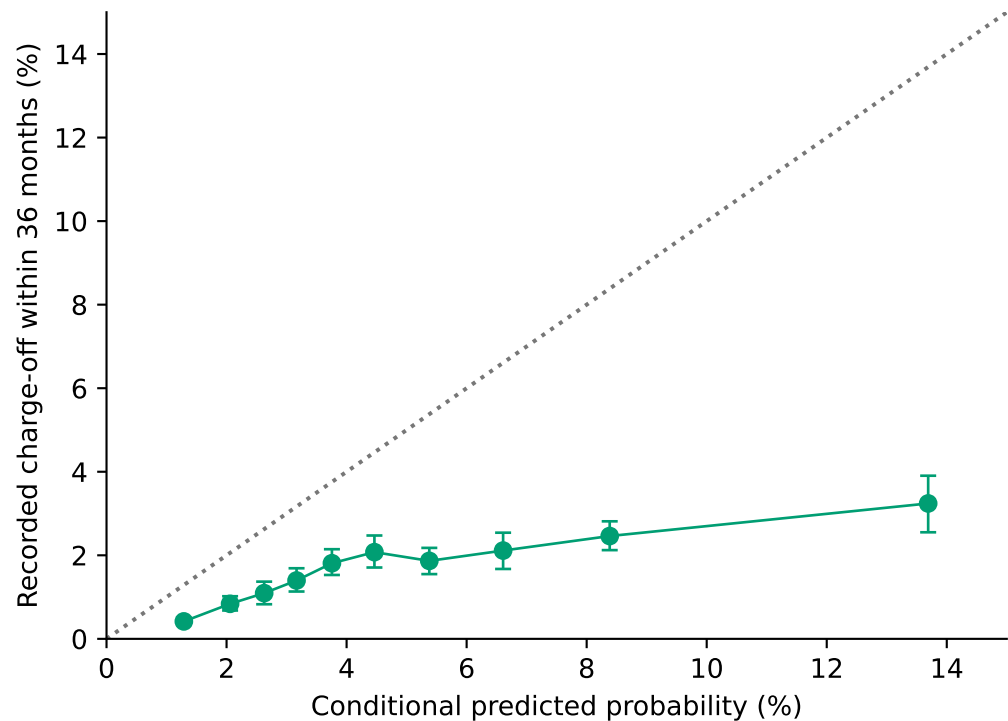


Macro LightGBM robustness: realized-path conditional calibration

Crisis 2007-2009



Later OOT 2011-2013



Future realized macros are scenario inputs. 95% state-cluster intervals; frozen fits.
This product uses FHFA data but is neither endorsed nor certified by FHFA.